



KENYATTA UNIVERSITY
EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE
(MECHANICAL ENGINEERING)

EMM 301: MECHANICS OF MACHINE I

1ST SEMESTER 2022/2023

TIME: 2 Hours

By Mr. Isaac Waire

INSTRUCTIONS:

1. Question ONE is Compulsory - 30 marks
2. Answer any TWO others (20 marks each)
3. You must have drawing equipment
4. Be neat and Tidy

Q1. (a) Define the following terms used with planer mechanisms:-

- (i) Mechanism
- (ii) Machine
- (iii) Kinematic pair
- (iv) Kinematic chain
- (v) Degree of freedom
- (vi) What is kinematic design? (6 marks)

b) Describe briefly, the following terms used in plane motion:

- (i) Translation, rectilinear translation, and curvilinear translation.
- (ii) Rotation (4 marks)

c) (i) Determine the positions of the revolute joints of the mechanisms

shown in figure Q1 (c), i. given that:

$\Phi = 60^\circ$ at the instant

AB= 110MM

BC= 160MM

$CD = 100\text{mm}$

$DE = 360\text{mm}$

Use the fixed X,Y frame as the reference frame.

(6 marks)

(ii) The four - bar linkage shown in figure Q1 C(ii) has the following link

lengths: $A_0A = 125\text{mm}$, $AB = 275$, $B_0B = 225\text{mm}$ and $A_0B_0 = 200\text{mm}$. For the position when $\Phi = -120^\circ$. Calculate the angular velocity of the output and of the coupler if the input angular velocity is 25 rad/s clockwise.

(5 marks)

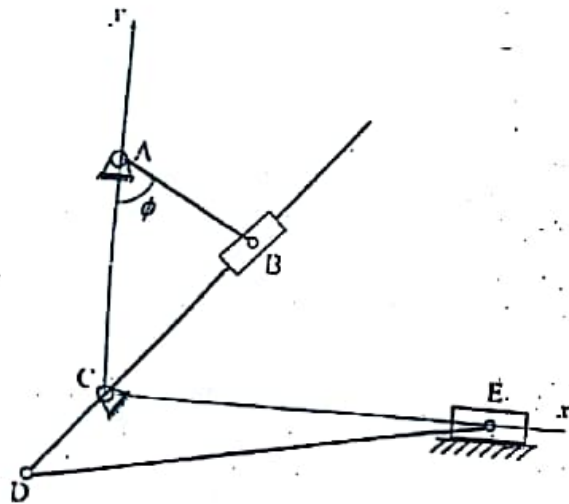


Fig. Q1(c) (i)

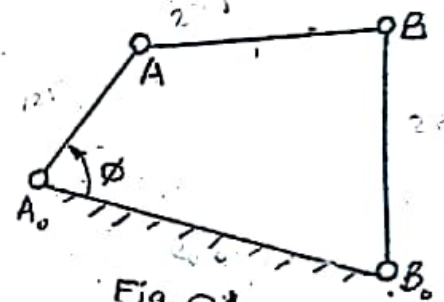


Fig. Q1(c) (ii)

- (d) (a) In a plane mechanism OABCE represented by figure Q2(C), the crank OA rotates at a constant speed of 150 rev/min anticlockwise. The connecting rod CBD moves about a pivot at B. Link CE is constrained to move horizontally at E. End C is pinned to an oscillating lever CE.

Take $AO = 60\text{mm}$, $BC = 62.5\text{mm}$, $AB = 163\text{mm}$, $CE = 200\text{mm}$, $\phi = 60^\circ$.

For the given configuration determine:-

- (i) The velocity of slider E
- (ii) The angular acceleration of the link DBC.

Use graphical method

(10 marks)

Q2. (a) Explain, giving examples of each, the following terms:-

- (i) Pairing elements
- (ii) A low pair
- (iii) A higher pair
- (iv) A link
- (v) Kinematic chain.

(5 marks)

- (b) (i) When does a constrained chain become a mechanism?

(1 mark)

- (ii) Sketch and name any three types of kinematic links.

(3 marks)

- (c) (i) The crank of a particular engine mechanism has a length of 35mm and the connecting rod a length of 124mm

- (a) Calculate the velocity of A pin-point, when the engine rotates at 2500 rev/min and the angle between the crank and the line of stroke is 45° , and also

(6 marks)

- (b) Calculate the velocity of the piston B.

(5 marks)

- (ii) The bar shown in figure Q3, C,ii, is confined to move in the horizontal and vertical slots A and B.

If the velocity of the slider block at A is 8m/s, determine the bar's angular velocity of block B at the instant when $\theta = 60^\circ$.

Use analytical techniques or instantaneous method

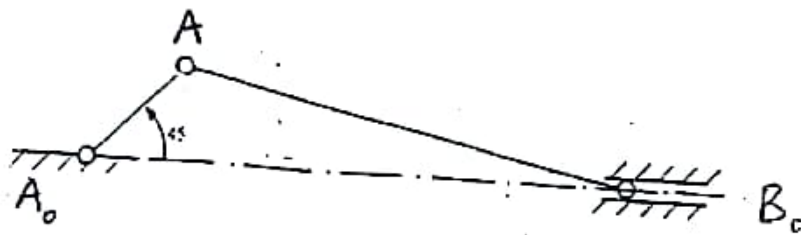


Fig. Q 2 (c) (i)

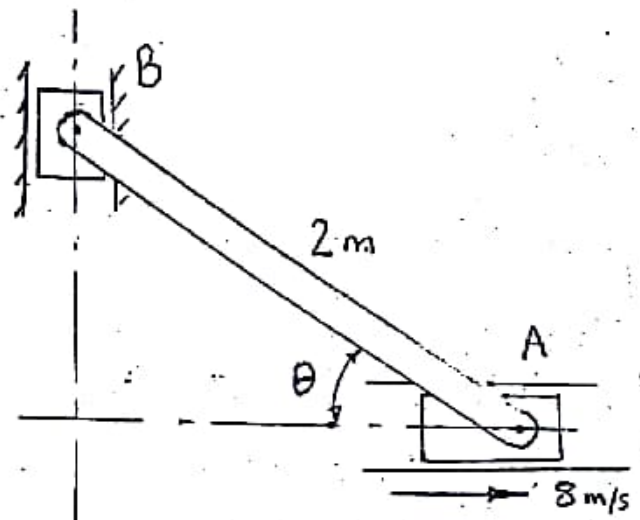


Fig 2(c) ii

Q3. (a) Derive the formula torque transmission $[T]$ for a cone clutch considering

- i) Uniform pressure
- ii) Uniform wear

Where W -Total Axial thrust transmitted to the clutch

r_1 and r_2 Outer and inner radius of friction surface respectively

α = Semi-angle of the cone

μ coefficient of friction between contact surfaces

(12 marks)

- b) A single dry plate clutch transmits 7.5 Kw at 900 r.p.m. The axial pressure is limited to 0.07 N/mm^2 If the coefficient of friction is 0.25 calculate:

- i) Mean radius
- ii) Face width of the friction lining assuming the ratio of the mean radius to the face width as 4

- i) Outer and inner radii of the clutch plate (8 marks)

Q4. (a) With an aid of well labeled diagram explain the following terms as used in spur gears

- i) Addendum
- ii) Dedendum
- ii) Circular pitch
- iii) Pitch circle diameter (4 marks)

- b) Derive the expression for the length of the path of contact for a wheel and pinion gear when in contact is given by:

$$\sqrt{(R_A)^2 - R^2 \cos^2 \phi} + \sqrt{(r_A)^2 - r^2 \cos^2 \phi} - (R + r) \sin \phi$$

Where R_A = radius of the addendum of circle of wheel

R = radius of pitch of the wheel

r_A = radius of addendum of circle of pinion

r = Radius of pitch circle of opinion (8 marks)

- c) A pinion having 30 teeth drives a gear having 80 teeth. The profile of the gear is involute with 20° pressure angle, 12mm module and 10mm addendum calculate the contact ratio (8 marks)

Q5. (a) State two advantages and two limitations of chain drive over belt drives. (2 marks)

Explain with sketches:

- i) Pitch of the chain
- ii) Pitch circle diameter of the chain sprocket (2 marks)

- b) Derive the relationship between pitch and pitch circle diameter for a chain drive. (4 marks)

- d) An open belt drive connects two pulleys 1.2 m and 0.5 m diameter, on parallel shafts 4 meters apart. The mass of the belt is 0.9 kg per meter length and the maximum tension is not to exceed 200 N. The coefficient of friction is 0.3. The 1.2 m pulley, which is the drive, runs at 200 r.p.m. due to belt slip on one of the pulleys, the velocity of the driven shaft is only 450 r.p.m. Calculate the
- Torque on each of the two shafts
 - Power transmitted
 - Power lost in friction
 - Efficiency of the drive.
- (12 marks)

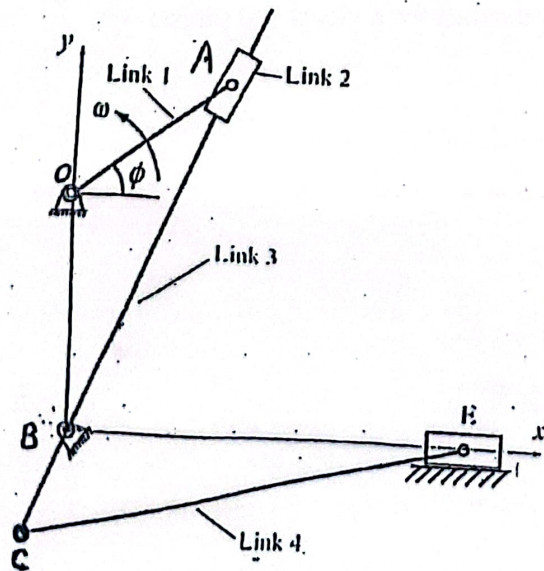


Fig. Q 1. (d)

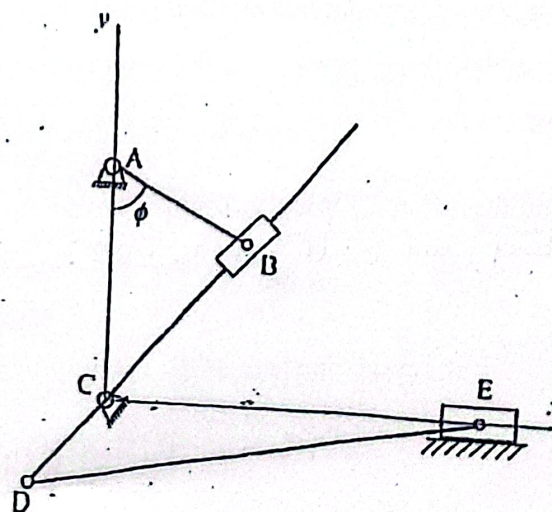


Fig. Q 2. (C)

Example 10.25. A single dry plate clutch transmits 7.5 kW at 900 r.p.m. The axial pressure is limited to 0.07 N/mm^2 . If the coefficient of friction is 0.25, find **1.** Mean radius and face width of the friction lining assuming the ratio of the mean radius to the face width as 4, and **2.** Outer and inner radii of the clutch plate.

Solution. Given : $P = 7.5 \text{ kW} = 7.5 \times 10^3 \text{ W}$; $N = 900 \text{ r.p.m}$ or $\omega = 2\pi \times 900/60 = 94.26 \text{ rad/s}$; $p = 0.07 \text{ N/mm}^2$; $\mu = 0.25$

1. Mean radius and face width of the friction lining

Let R = Mean radius of the friction lining in mm, and

w = Face width of the friction lining in mm,

Ratio of mean radius to the face width,

$$R/w = 4 \quad \dots(\text{Given})$$

We know that the area of friction faces,

$$A = 2\pi R.w$$

$\therefore$ Normal or the axial force acting on the friction faces,

$$W = A \times p = 2\pi R.w.p$$

We know that torque transmitted (considering uniform wear),

$$\begin{aligned} T &= n.\mu.W.R = n.\mu (2\pi R.w.p) R \\ &= n.\mu \left(2\pi R \times \frac{R}{4} \times p \right) R = \frac{\pi}{2} \times n.\mu.p.R^3 \quad \dots(\because w = R/4) \end{aligned}$$

$$= \frac{\pi}{2} \times 2 \times 0.25 \times 0.07 R^3 = 0.055 R^3 \text{ N-mm} \quad \dots(i)$$

$\dots(\because n = 2, \text{ for single plate clutch})$

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We also know that power transmitted (P),

$$7.5 \times 10^3 = T.\omega = T \times 94.26$$

$$\therefore T = 7.5 \times 10^3 / 94.26 = 79.56 \text{ N-m} = 79.56 \times 10^3 \text{ N-mm} \quad \dots(ii)$$

From equations (i) and (ii),

$$R^3 = 79.56 \times 10^3 / 0.055 = 1446.5 \times 10^3 \text{ or } R = 113 \text{ mm Ans.}$$

and $w = R/4 = 113/4 = 28.25 \text{ mm Ans.}$

2. Outer and inner radii of the clutch plate

Let r_1 and r_2 = Outer and inner radii of the clutch plate respectively.

Since the width of the clutch plate is equal to the difference of the outer and inner radii, therefore

$$w = r_1 - r_2 = 28.25 \text{ mm} \quad \dots(iii)$$

Also for uniform wear, the mean radius of the clutch plate,

$$R = \frac{r_1 + r_2}{2} \quad \text{or} \quad r_1 + r_2 = 2R = 2 \times 113 = 226 \text{ mm} \quad \dots(iv)$$

From equations (iii) and (iv),

$$r_1 = 127.125 \text{ mm ; and } r_2 = 98.875 \text{ Ans.}$$

Example 12.3. A pinion having 30 teeth drives a gear having 80 teeth. The profile of the gears is involute with 20° pressure angle, 12 mm module and 10 mm addendum. Find the length of path of contact, arc of contact and the contact ratio.

Solution. Given : $t = 30$; $T = 80$; $\phi = 20^\circ$;
 $m = 12$ mm ; Addendum = 10 mm

Length of path of contact

We know that pitch circle radius of pinion,

$$r = m.t / 2 = 12 \times 30 / 2 = 180 \text{ mm}$$

and pitch circle radius of gear,

$$R = m.T / 2 = 12 \times 80 / 2 = 480 \text{ mm}$$

$\therefore$ Radius of addendum circle of pinion,

$$r_A = r + \text{Addendum} = 180 + 10 = 190 \text{ mm}$$

and radius of addendum circle of gear,

$$R_A = R + \text{Addendum} = 480 + 10 = 490 \text{ mm}$$

We know that length of the path of approach,

$$KP = \sqrt{(R_A)^2 - R^2 \cos^2 \phi} - R \sin \phi \quad \dots(\text{Refer Fig. 12.11})$$

$$= \sqrt{(490)^2 - (480)^2 \cos^2 20^\circ} - 480 \sin 20^\circ = 191.5 - 164.2 = 27.3 \text{ mm}$$

and length of the path of recess,

$$PL = \sqrt{(r_A)^2 - r^2 \cos^2 \phi} - r \sin \phi$$

$$= \sqrt{(190)^2 - (180)^2 \cos^2 20^\circ} - 180 \sin 20^\circ = 86.6 - 61.6 = 25 \text{ mm}$$

We know that length of path of contact,

$$KL = KP + PL = 27.3 + 25 = 52.3 \text{ mm} \quad \text{Ans.}$$



Worm.

Length of arc of contact

We know that length of arc of contact

$$= \frac{\text{Length of path of contact}}{\cos \phi} = \frac{52.3}{\cos 20^\circ} = 55.66 \text{ mm} \quad \text{Ans.}$$

Contact ratio

We know that circular pitch,

$$p_c = \pi.m = \pi \times 12 = 37.7 \text{ mm}$$

$$\therefore \text{Contact ratio} = \frac{\text{Length of arc of contact}}{p_c} = \frac{55.66}{37.7} = 1.5 \text{ say } 2 \quad \text{Ans.}$$

Example 11.11. An open belt drive connects two pulleys 1.2 m and 0.5 m diameter, on parallel shafts 4 metres apart. The mass of the belt is 0.9 kg per metre length and the maximum tension is not to exceed 2000 N. The coefficient of friction is 0.3. The 1.2 m pulley, which is the driver, runs at 200 r.p.m. Due to belt slip on one of the pulleys, the velocity of the driven shaft is only 450 r.p.m. Calculate the torque on each of the two shafts, the power transmitted, and power lost in friction. What is the efficiency of the drive ?

Solution. Given : $d_1 = 1.2$ m or $r_1 = 0.6$ m ; $d_2 = 0.5$ m or $r_2 = 0.25$ m ; $x = 4$ m ; $m = 0.9$ kg/m ; $T = 2000$ N ; $\mu = 0.3$; $N_1 = 200$ r.p.m. ; $N_2 = 450$ r.p.m.

We know that velocity of the belt,

$$v = \frac{\pi d_1 N_1}{60} = \frac{\pi \times 1.2 \times 200}{60} = 12.57 \text{ m/s}$$

and centrifugal tension, $T_C = m.v^2 = 0.9 (12.57)^2 = 142$ N

$\therefore$ Tension in the tight side of the belt,

$$T_1 = T - T_C = 2000 - 142 = 1858 \text{ N}$$

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We know that for an open belt drive,

$$\sin \alpha = \frac{r_1 - r_2}{x} = \frac{0.6 - 0.25}{4} = 0.0875 \text{ or } \alpha = 5.02^\circ$$

$\therefore$ Angle of lap on the smaller pulley,

$$\begin{aligned} \theta &= 180^\circ - 2\alpha = 180^\circ - 2 \times 5.02^\circ = 169.96^\circ \\ &= 169.96 \times \pi / 180 = 2.967 \text{ rad} \end{aligned}$$

Let T_2 = Tension in the slack side of the belt.

We know that

$$2.3 \log \left(\frac{T_1}{T_2} \right) = \mu \cdot \theta = 0.3 \times 2.967 = 0.8901$$

$$\log \left(\frac{T_1}{T_2} \right) = \frac{0.8901}{2.3} = 0.387 \text{ or } \frac{T_1}{T_2} = 2.438$$

...(Taking antilog of 0.387)

$$\therefore T_2 = \frac{T_1}{2.438} = \frac{1858}{2.438} = 762 \text{ N}$$

Torque on the shaft of larger pulley

We know that torque on the shaft of larger pulley,

$$T_L = (T_1 - T_2) r_1 = (1858 - 762) 0.6 = 657.6 \text{ N-m} \quad \text{Ans.}$$

Torque on the shaft of smaller pulley

We know that torque on the shaft of smaller pulley,

$$T_S = (T_1 - T_2) r_2 = (1858 - 762) 0.25 = 274 \text{ N-m} \quad \text{Ans.}$$

Power transmitted

We know that the power transmitted,

$$\begin{aligned} P &= (T_1 - T_2) v = (1858 - 762) 12.57 = 13780 \text{ W} \\ &= 13.78 \text{ kW} \quad \text{Ans.} \end{aligned}$$

Power lost in friction

We know that input power,

$$P_1 = \frac{T_L \times 2\pi N_1}{60} = \frac{657.6 \times 2\pi \times 200}{60} = 13780 \text{ W} = 13.78 \text{ kW}$$

$$\text{and output power, } P_2 = \frac{T_S \times 2\pi N_2}{60} = \frac{274 \times 2\pi \times 450}{60} = 12910 \text{ W} = 12.91 \text{ kW}$$

$$\therefore \text{Power lost in friction} = P_1 - P_2 = 13.78 - 12.91 = 0.87 \text{ kW} \quad \text{Ans.}$$

Efficiency of the drive

We know that efficiency of the drive,

$$\eta = \frac{\text{Output power}}{\text{Input power}} = \frac{12.91}{13.78} = 0.937 \text{ or } 93.7\% \quad \text{Ans.}$$